Benjamin Duffaut

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Education

Bachelors of Science in Computer Science August 2023 – May 2027
University of Central Florida, Burnett Honors College | Provost Scholar *Orlando, FL*

Relevant Coursework: Data Structures and Algorithms, Systems Software, Compilers, Neural Networks and Deep Learning, Front-end Development

Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, HTML/CSS, Bash
Frameworks: React, Vue.js, Node.js, Express.js, Flutter
Testing: Vitest, Jest, Playwright
Tools: MongoDB, MySQL, Git, Docker, Jira, Figma, Postman

Experience

Software Engineer Intern June 2026 – October 2026
Reveal Security *New York, NY (Remote)*

- Contributed 5+ shipped features to the production codebase, working directly with the CTO and Senior Director of Product on Vue.js frontend and supporting backend work.
- Built a fully custom, componentized dashboard using GridStack.js with widget-level data caching, so adding, removing, and rearranging widgets didn't degrade performance; delivered real-time alerts with direct navigation into detailed views.
- Wrote frontend unit and browser tests with Vitest and Playwright and backend API tests with Jest, using Mockoon for mock data-ingestion APIs and pgAdmin for test data seeding.

Frontend Developer Intern August 2025 – May 2026
Compete Co. *Orlando, FL*

- Developed and implemented 6+ core frontend features of the Compete mobile app using React Native with Jest-covered unit tests, improving UI consistency and user engagement across the platform.
- Collaborated in a 6-member Agile team using Jira for sprint planning, building and optimizing 5+ responsive UIs with designers and backend engineers to hit design specs and deadlines, contributing to on-time delivery of 100% of sprint goals.

Projects

CourtSense – AI Tennis Analysis System March 2026 – June 2026

- Built a multi-modal AI tennis analysis system in PyTorch, integrating five neural network architectures: a CNN classifier (99.8% val accuracy), a court keypoint detector (0.04px val error), a single-stage object detector (F1 0.894), a ball trajectory LSTM, and a VLM.
- Designed a modular architecture with independently trainable, swappable inference interfaces, implementing the full training pipeline from scratch, including custom loss functions and hardware-aware device selection across MPS, CUDA, and CPU.

Personalized News Platform September 2025 – November 2025

- Designed and built a personalized news platform that curates content based on user profiles and interests gathered during onboarding, integrating a third-party news API with an LLM-powered filtering system for contextually relevant articles.
- Implemented optimized API handling, state management, and database pipelines to support dynamic, user-specific content across web and mobile clients.

Activities

UCF Club Tennis | *National Team Member* August 2023 – Present

- Competed as a member of the UCF Club Tennis national team for three consecutive years, qualifying for and competing at USTA/ITA Club Tennis Nationals, finishing 12th in the nation.